

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - V

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Class :SYIT	Batch : 00
Date of Assignment :12-08-2026	Date/Time of Submission :12-08-2026

Part – A Theory

5a https://www.w3schools.com/python/pandas/ref_df_filter.asp

5b https://www.w3schools.com/python/module_statistics.asp

5c <https://www.geeksforgeeks.org/python/creating-a-pandas-series-from-dictionary/>

5d <https://www.geeksforgeeks.org/pandas/boolean-indexing-in-pandas/>

PART – B Practical Questions:

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

Code:

```
import pandas as pd
```

```
data = {  
    "Name": ["Aniket", "Maruti", "Paneer", "Shubhangi"],  
    "Age": [50, 40, 30, 20],  
    "Qualified": [True, False, False, True]  
}  
df = pd.DataFrame(data)
```

```
print(df)
```

```
print("Aniket")
```

Output:

```
\Python314\MVLU\Chaman.py  
      Name  Age  Qualified  
0   Aniket   50      True  
1  Maruti   40     False  
2  Paneer   30     False  
3 Shubhangi  20      True  
Aniket
```

5b. Write a python code to show statistical information on given data set.

Code:

```
import statistics
```

```

print("Stats Harmonic Mean: ",statistics.harmonic_mean([40, 60, 80]))
print("Stats Harmonic Mean: ",statistics.harmonic_mean([10, 30, 50, 70, 90]))
print("Stats Mean: ",statistics.mean([40, 60, 80]))
print("Stats Mean: ",statistics.mean([10, 30, 50, 70, 90]))
print("Stats Median: ",statistics.median([40, 60, 80]))
print("Stats Median: ",statistics.median([10, 30, 50, 70, 90]))
print("Stats Median Grouped: ",statistics.median_grouped([40, 60, 80]))
print("Stats Median Grouped : ",statistics.median_grouped([10, 30, 50, 70, 90]))
print("Stats Mode: ",statistics.mode([40, 60, 80]))
print("Stats Mode: ",statistics.mode([10, 30, 50, 70, 90]))
print("Aniket")

```

Output:

```

\Python314\MVLU\Chaman.py
Stats Harmonic Mean:  55.38461538461538
Stats Harmonic Mean:  27.97513321492007
Stats Mean:  60
Stats Mean:  50
Stats Median:  60
Stats Median:  50
Stats Median Grouped:  60.0
Stats Median Grouped :  50.0
Stats Mode:  40
Stats Mode:  10
Aniket

```

5c. Write a python code to create pandas series from dictionaries.

Code:

```
import pandas as pd
```

```
dictionary = {'A': 10, 'B': 20, 'C': 30}
```

```
series = pd.Series(dictionary)
```

```
print(series)
```

```
print("Aniket")
```

Output:

```
A      10
B      20
C      30
dtype: int64
Aniket
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

Code:

```
import pandas as pd
```

```
dict = {'Name':["Suparna", "Roshan", "Amit", "Shubhangi"],
        'Degree': ["MBA", "BCA", "M.Tech", "MBA"],
        'Score':[90, 40, 80, 98]}
```

```
df = pd.DataFrame(dict, index = [True, False, True, False])
```

```
print(df)
```

```
print("Aniket")
```

Output:

```

Python314\MVLU\Chaman.py
Name Degree Score
True Suparna MBA 90
False Roshan BCA 40
True Amit M.Tech 80
False Shubhangi MBA 98
Aniket

```

PART – C Curiosity Question: